Sparse Decomposition Captures Neural Activity Better than Discrete State Models

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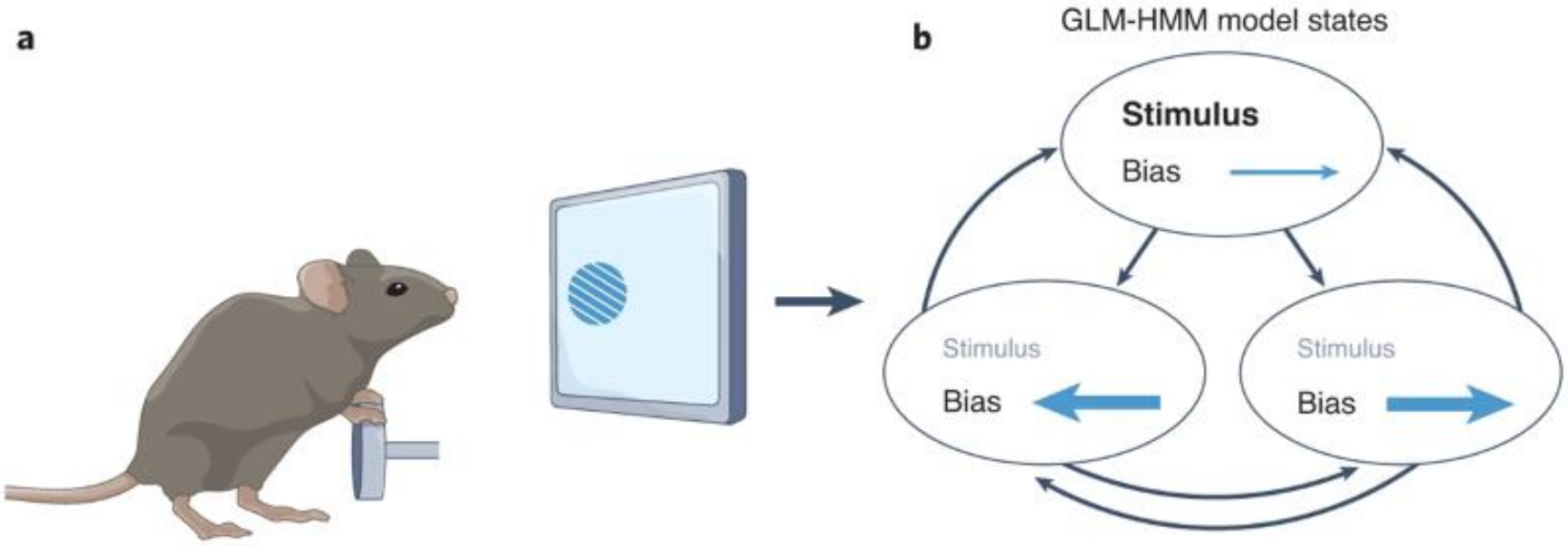
imbizo / ɪm'bi:zo / (Xhosa) :
A gathering to share knowledge

GLM-HMM Approach

- Discrete state model $z \in \{1 \ldots K\}$



Research Question

- What if **"brain states"** aren't switches, but recipes?

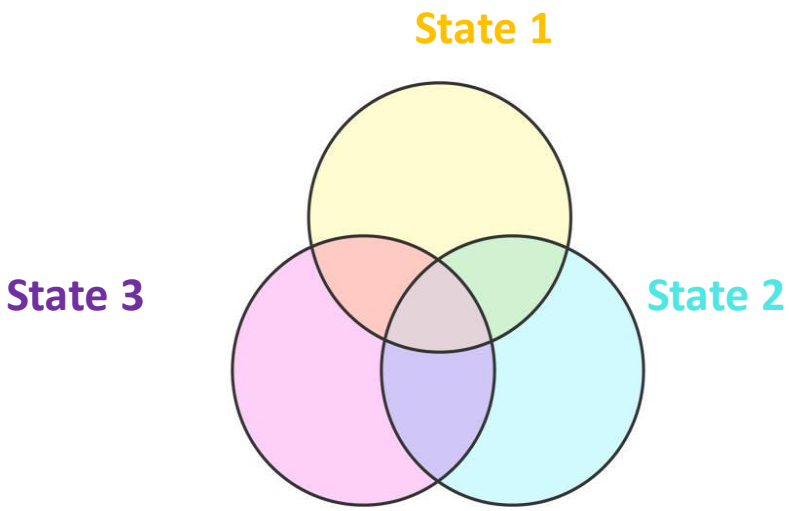
Do neural activities switch between discrete state or blends continuous motifs?

Why Sparsity Matters

- Neurons respond to limited sets of features at any moment – not all motifs fire simultaneously
- Sparse codes are easier to interpret
- Without sparsity constraint, infinite solutions exist

Proposed Approach: Sparse Mixture Model

- Continuous sparse states
- Soft mixing of states



$$\alpha \in \mathbb{R}^K$$

Neuroscience shows 'mixed selectivity' — neurons respond to combinations of task-relevant features simultaneously *(Rigotti et al., 2013)*

Poisson Generative Model

$$y_{n,i} \sim \text{Poisson}(r_{n,i})$$

$$\text{where} \quad r_{n,i} = \sum_k \lambda_{nk} \cdot \alpha_{ik}$$

λ - Basis Patterns

Shape: $(N \times K)$
N = neurons
K = motifs
Fixed firing rates

α - Mixing Weights

Shape: $(K \times M)$
M = time samples
SPARSE: $\|\alpha\|_0 \leq S$
Normalized:
$\sum_k \alpha_k = 1$

y - Observations

Shape: $(N \times M)$
Spike counts
Poisson distributed
Observable data

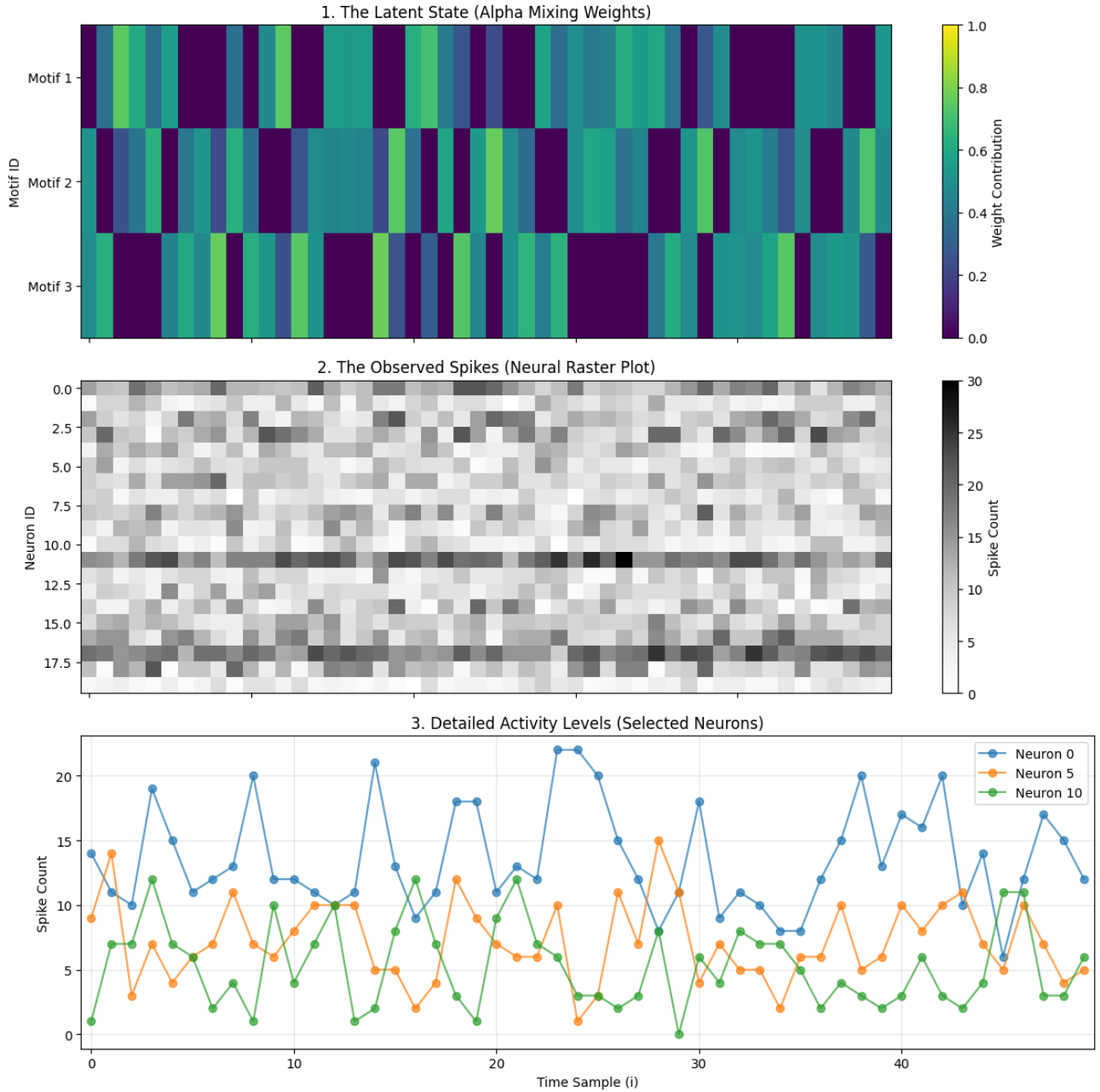
Synthetic Data Generation

1. Initialize λ — $\lambda_k \sim \text{Uniform}(0, \lambda_{\max})^N$
2. Sparse Support — **Select S active motifs** $S_i \subseteq \{1, \ldots, K\}, \quad |S_i| = 2$
3. Mix weights — **Sample & normalize** $\alpha_{ik} \sim \text{Uniform}, \quad \sum_k \alpha_k = 1$
4. Compute rates — **Linear combination** $r_i = \sum_k \alpha_{ik} \cdot \lambda_k$
5. Sample spikes — **Poisson observation** $y_n \sim \text{Poisson}(r_n)$

Configuration

N = 20 neurons
K = 3 motifs
M = 50 samples
S = 2 active/sample

$\lambda_{\max}$ = 20 Hz
$\alpha_{\min}$ = 0.2
$\alpha_{\max}$ = 1.0

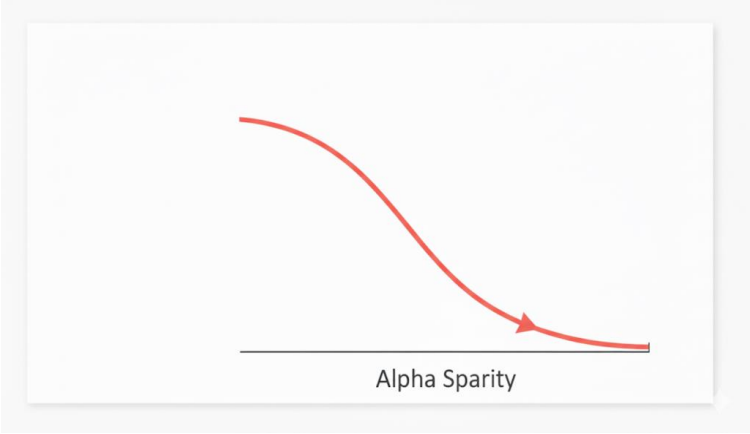

1. The Latent State (Alpha Mixing Weights)
Motif ID
Motif 1
Motif 2
Motif 3
Weight Contribution
1.0
0.8
0.6
0.4
0.2
0.0
2. The Observed Spikes (Neural Raster Plot)
Neuron ID
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Spike Count
30
25
20
15
10
5
0
3. Detailed Activity Levels (Selected Neurons)
Neuron 0
Neuron 5
Neuron 10
Spike Count
Time Sample (i)
0
10
20
30
40

Sparse Poisson Factor Model

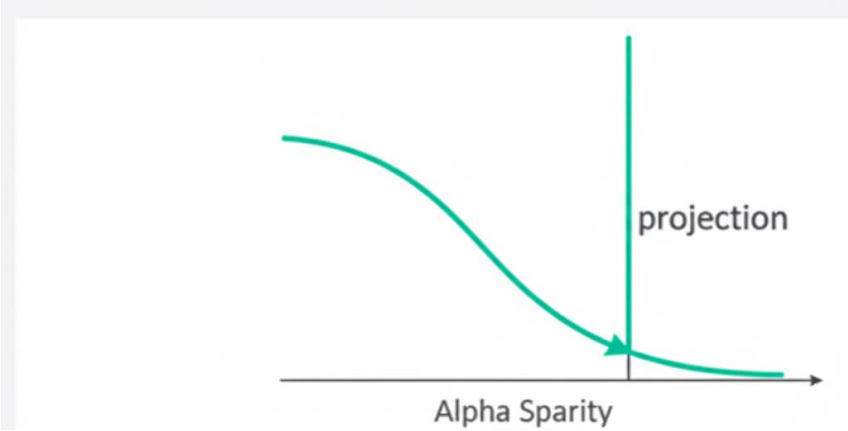
Objective: Minimize Poisson Negative Log-Likelihood

$$\mathcal{L}(\lambda, \alpha) = \sum_i \sum_n \left[r_{ni} - y_{ni} \cdot \log(r_{ni}) \right] + \beta \cdot \|\alpha\|_1$$

L1 Penalty



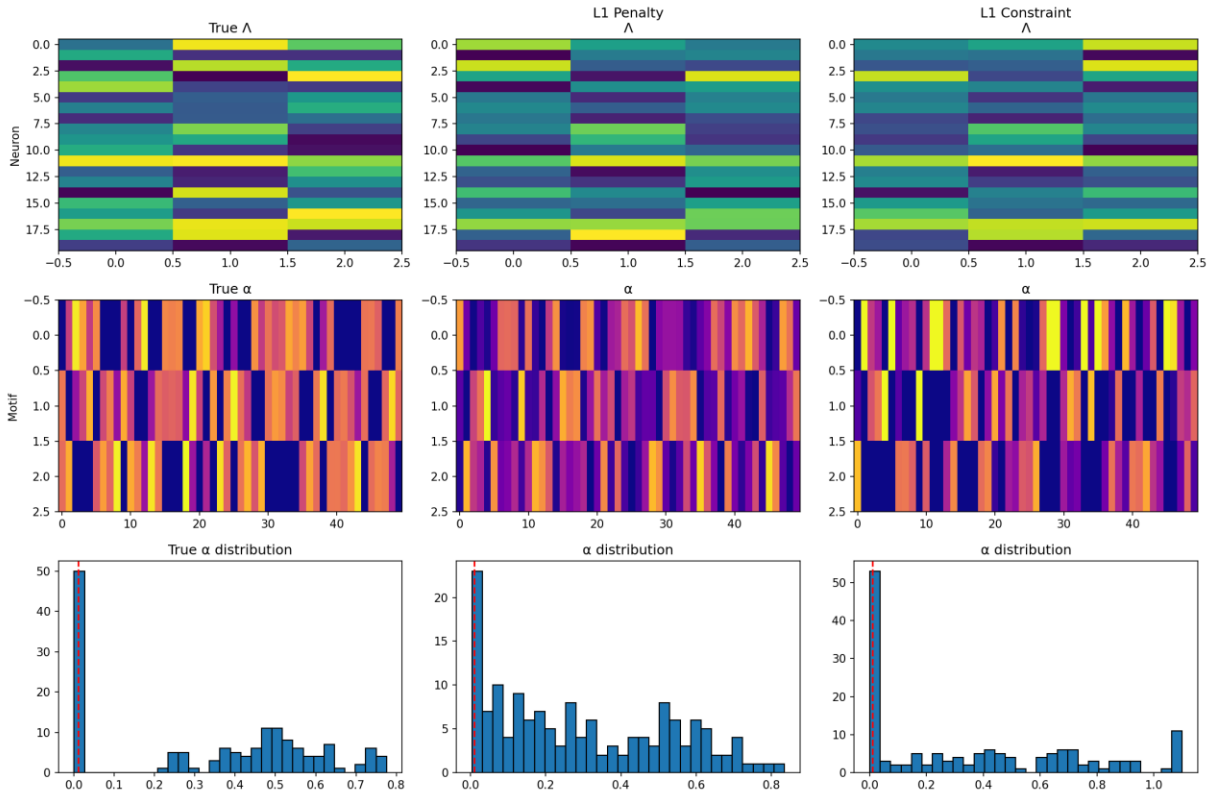
Projected Gradient



Key Results

Table 1: Comparison of Sparsity Recovery Across Optimization Methods

Method	Sparsity %	Exact Zeros	Alignment Score (r)
Standard Poisson NLL	42.15%	0.00%	0.88
L1 Regularization	78.40%	12.30%	0.92
Projected Gradient	**91.60%**	**89.45%**	**0.96**
Ground Truth	*92.00%*	*92.00%*	*1.00*



Parameter Recovery Correlations

Recovery Correlations (with Ground Truth)

0.96	0.82	91
Reconstuction r(true, fitted)	Basis Patterns λ correlation	Mixing Weights α correlation

Conclusion

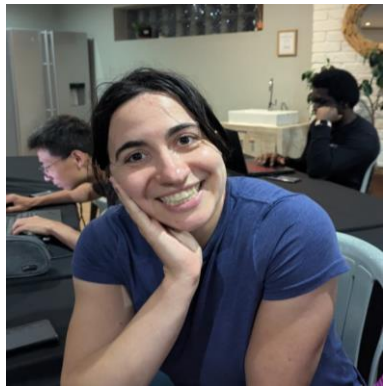
- Sparse continuous latent models can successfully decompose neural population activity into interpretable "basis motifs" with known mixing dynamics

- The projected gradient method with L1 constraint accurately recovers the true sparse structure outperforming soft penalty approaches

Future Works

- Compare with GLM-HMM
- Experiment on a real data; Steinmetz 2019 dataset

Acknowledgement



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Thank You

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